

Introduction to Circles

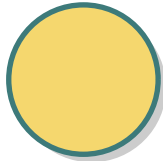
A fun, illustrated lesson for curious kids!



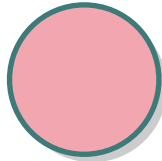
What is a Circle?

A **circle** is a round shape where **every point on the edge is exactly the same distance from the centre**. It has no corners and no straight sides — just one perfectly smooth, continuous curve!

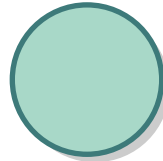
Look around you — circles are everywhere:



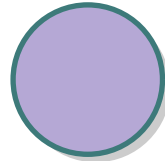
Plate



Clock



Wheel

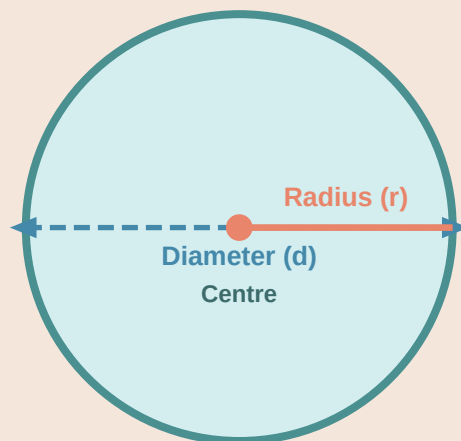


Bangle



Parts of a Circle

Circumference ($C = \pi \times d$)



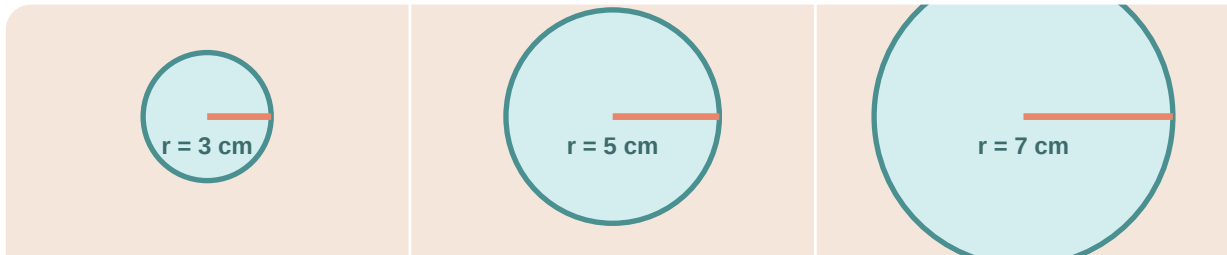
Part	What it means	Fun way to remember
Centre	The exact middle point of a circle	The dot in the bull's-eye!
Radius (r)	Distance from the centre to the edge	Half of a pizza slice
Diameter (d)	Distance all the way across through the centre	Two radii back-to-back

Circumference	The distance all the way around the circle	Ribbon wrapped around a cookie
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What is the Radius?

The **radius** is the distance from the **centre** of a circle to any point on its **edge**. Because a circle is perfectly round, *every radius of the same circle is exactly the same length*.

Think of a pizza slice — the straight edge from the very middle of the pizza out to the crust is the radius! ■



What is the Diameter?

The **diameter** is the distance all the way **across** a circle, passing straight through the centre. It is always exactly **twice the radius**.

$$\text{Diameter} = 2 \times \text{Radius} \quad (d = 2r)$$

So if the radius is 5 cm, the diameter is 10 cm!

	Radius	Diameter
Measures	Centre → Edge	Edge → Edge (through centre)
Formula	$r = d \div 2$	$d = 2 \times r$
Example	$r = 5 \text{ cm}$	$d = 10 \text{ cm}$
Real-world	Spoke of a bicycle wheel	Full width of a round plate

What is the Circumference?

The **circumference** is the total distance all the way **around** the circle — like measuring how far you would walk if you jogged one full lap around a circular park.

- Imagine wrapping a ribbon around a circular cookie — the length of that ribbon is the circumference!

The Circumference Formula

$$C = \pi \times d$$

where **C** = Circumference, π (Pi) ≈ 3.14 , **d** = Diameter
You can also write it as **C** = $2 \times \pi \times r$ (using the radius instead).



Step-by-Step Example

Step	What to do	Working
1	Write down the diameter	$d = 10 \text{ cm}$
2	Write down Pi	$\pi \approx 3.14$
3	Multiply $\pi \times d$	3.14×10
4	Calculate the answer	C = 31.4 cm ✓

Fun Circle Facts! ■



No Corners!

A circle is the only 2D shape with no corners and no straight sides at all.



Pi goes on forever!

Pi ($\pi = 3.14159\dots$) is an *irrational number* — its decimal digits never repeat and never end!



Wheels are circles!

Circular wheels roll smoothly because every point on the rim stays the same distance from the axle (the centre).



Circles in nature

The Sun, the Moon, ripples in water, and the pupils of your eyes are all approximately circular.



Quick Reference Card ■

Term	Symbol	Formula / Definition
Radius	r	Distance from centre to edge
Diameter	d	$d = 2 \times r$ (edge to edge through centre)
Circumference	C	$C = \pi \times d = 2 \times \pi \times r$
Pi	π	≈ 3.14 (or $22/7$)
Area	A	$A = \pi \times r^2$ (bonus formula!)

Practice Time! ➡

Question 1

— What is the radius of a circle with diameter 14 cm?

A) 14 cm

B) 28 cm

C) 7 cm ✓

D) 3.14 cm

Question 2

— Which formula gives the circumference?

A) $C = r \times r$

B) $C = \pi \times d$ ✓

C) $C = d + r$

D) $C = 2 + r$

Question 3

— If radius = 6 cm, what is the diameter?

A) 3 cm

B) 6 cm

C) 12 cm ✓

D) 18.84 cm

Question 4

= Pi (π) is approximately equal to ...

A) 2.71

B) 1.41

C) 3.14 ✓

D) 4.00



Great job learning about Circles! ■

You now know the **centre**, **radius**, **diameter**, and **circumference** of a circle — plus the magic of **Pi (π)**. Keep looking for circles all around you!